Optimising using corner points prin[redacted]

1

a

i 14 **ii** 39 **iii** 54 **iv** 44

b The maximum value of the objective function is 54 at the point $(6, 10)$.

c The minimum value of the objective function is 14 at the point $(2, 2)$.

2

a

i The maximum value of the objective function is 48 at the point $(6, 9)$.

ii The minimum value of the objective function is 14 at the point $(2, 2)$.

b

i The maximum value of the objective function is 87 at the point $(9, 10)$.

ii The minimum value of the objective function is 15 at the point $(1, 2)$.

c

i The maximum value of the objective function is 51 at the point $(8, 9)$.

ii The minimum value of the objective function is 12 at the point $(2, 2)$.

d

i The maximum value of the objective function is 90 at the point $(3, 9)$.

ii The minimum value of the objective function is 0 at the point $(3, 0)$.

e

i The maximum value of the objective function is 35 at the point $(9, 5)$

ii The minimum value of the objective function is -10 at the point $(2, 10)$.

3 The maximum value of the objective function T is 279 at the point $(0, 9)$.

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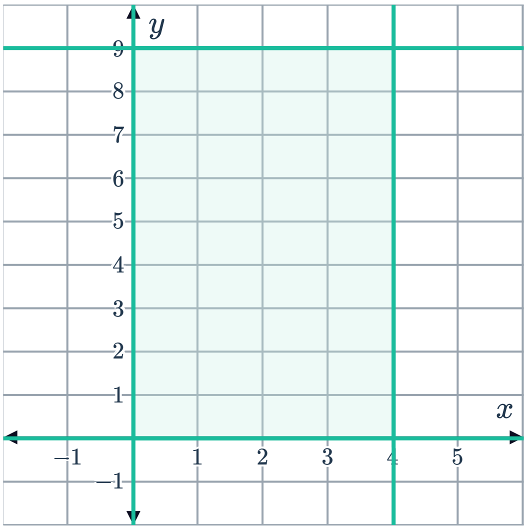
4

a

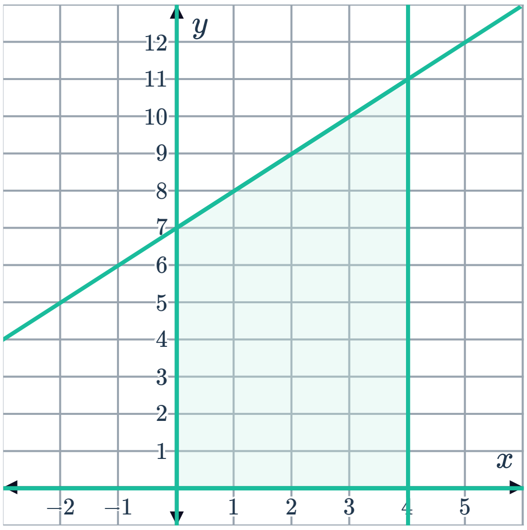
i



ii $(0,0),(4,0),(4,9),(0,9)$

b

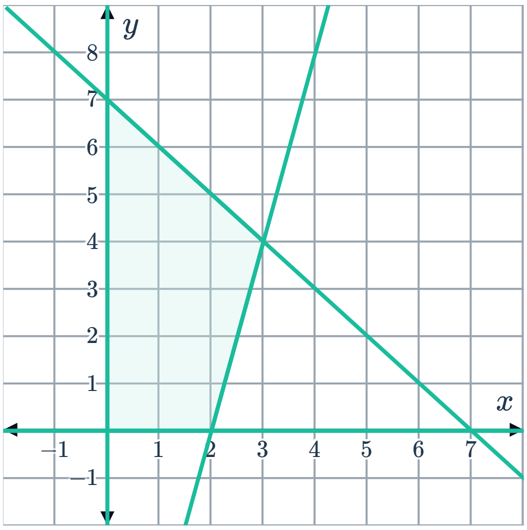
i



ii $(0,0),(4,0),(4,11),(0,7)$

c

i



ii $(0,0),(2,0),(3,4),(0,7)$

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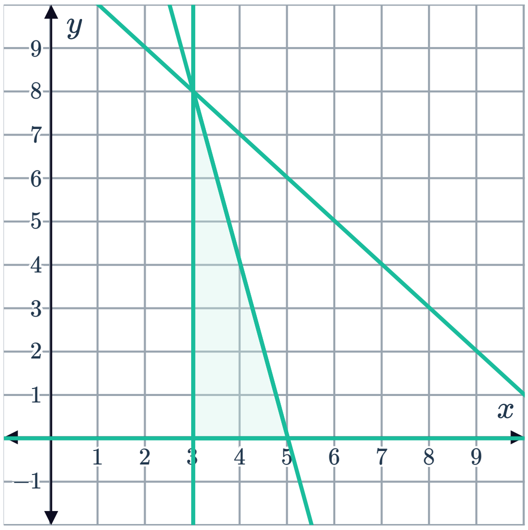
d

i

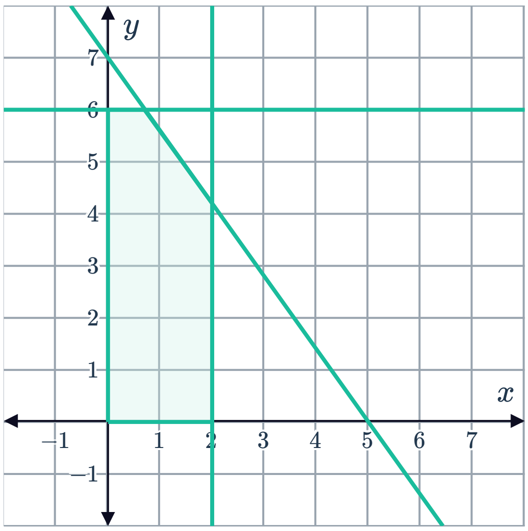


ii $(3,0)\,,(5,0)\,,(3,8)$

5

a

i



ii The maximum is 116 at the point $(0,6)$.

iii The minimmum is 0 at the point $(2,0)$

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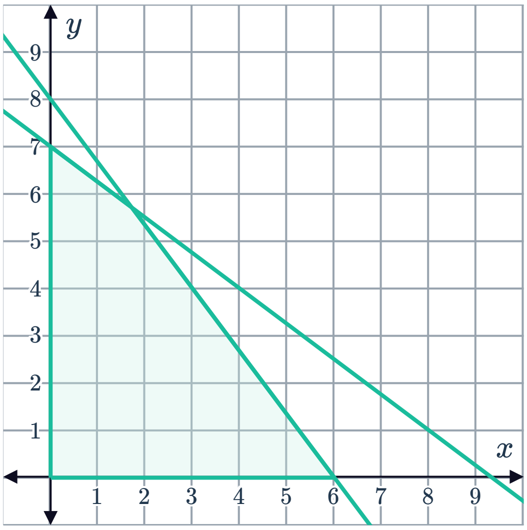
b

i



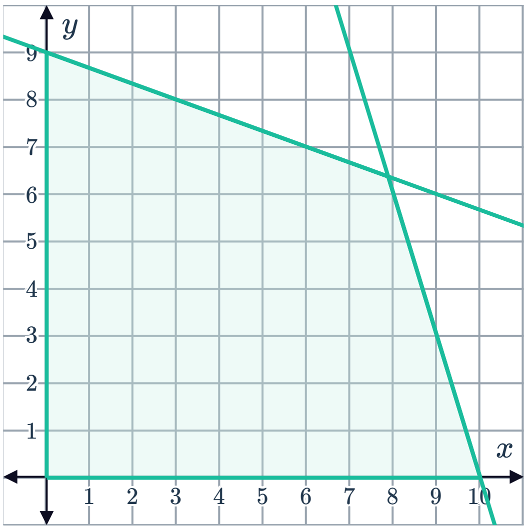
ii The maximum is 322 at the point $(0, 7)$.

iii The minimum is 0 at the point $(0, 0)$.

6

a

i

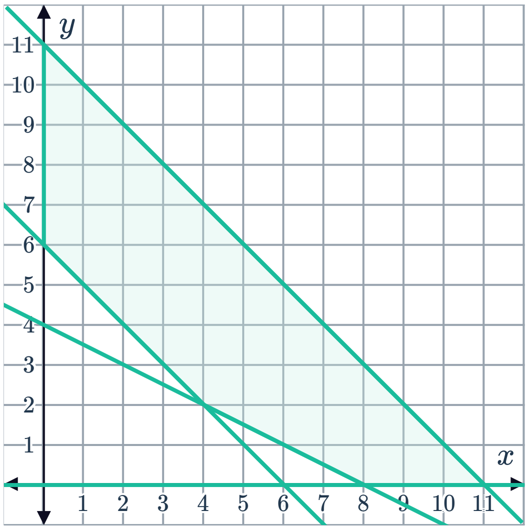


ii The maximum value of P is 60 at $(10, 0)$.

iii No solution provided.

b

i

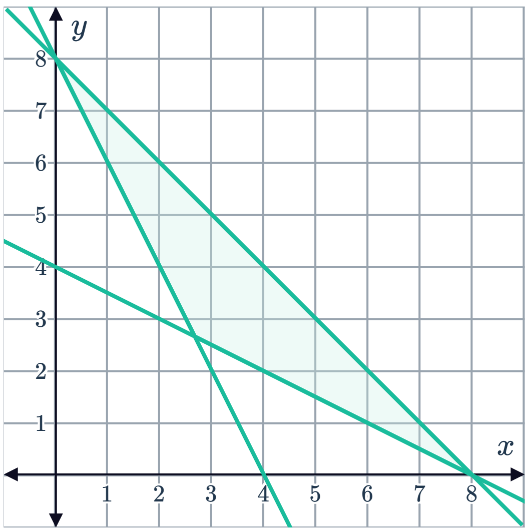


ii The maximum value of z is 242 at $(11, 0)$.

iii No solution provided

c

i



ii The maximum value of z is 112 at $(0, 8)$.

iii No solution provided

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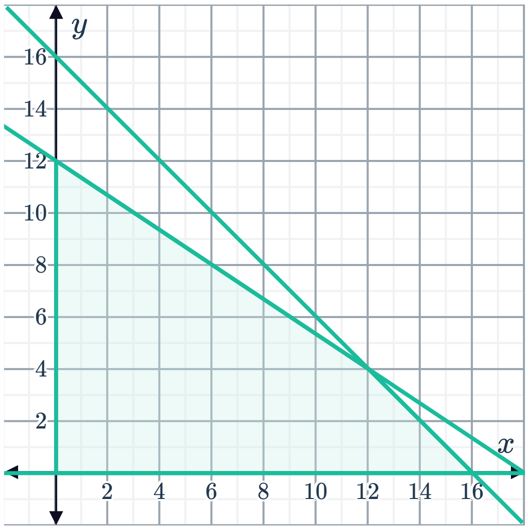
Applications

7

a $x \geq 0, y \geq 0, 2x + 3y \leq 36,\ 2x + 2y \leq 32$

b



c $(0,0), (0,12), (12,4), (16,0)$

d $P = 240x + 280y$

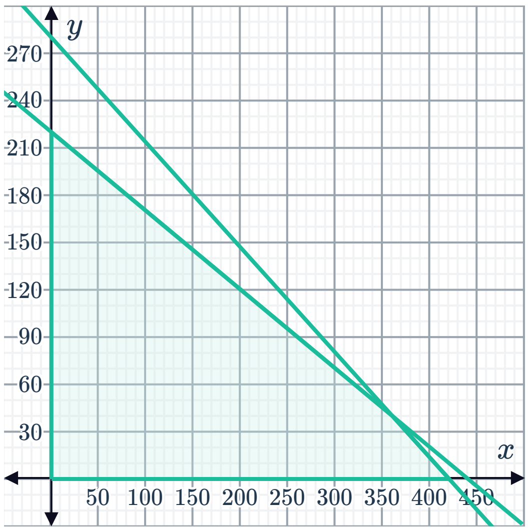
e $(12,4)$

f $\$4000$

8

a $x \geq 0, y \geq 0, 8x + 16y \leq 3520$ and $8x + 12y \leq 3360$

b



c $(0,0), (0,220), (360,40), (420,0)$

d $P = 15x + 26y$

e $(360,40)$

f $\$6640$

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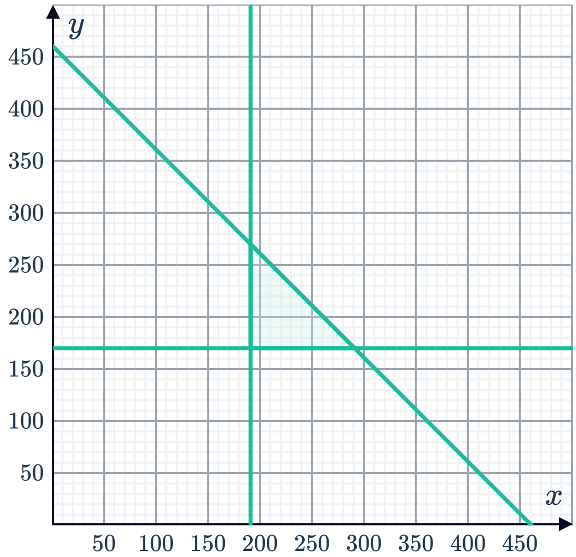
9

a $x \geq 190, y \geq 170$ and $x + y \leq 460$

b



c $P = 1.70x + 1.00y$

d

Vertex	Profit ($)
$(190, 170)$	493
$(190, 270)$	593
$(290, 170)$	663

e $663

f 290 beef and 170 chicken burgers.

g 190 beef and 270 chicken burgers.

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